The Role of the Liver in Excretion

Exam code: 0610 & 0970

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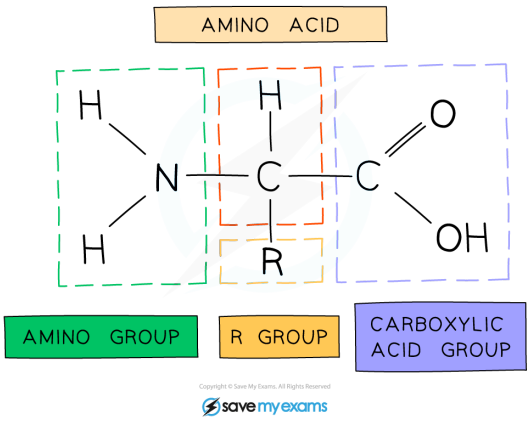
Excretion by Deamination of Amino Acids: Extended

- Many digested food molecules absorbed into the blood in the small intestine are carried to the liver for **assimilation** (when food molecules are converted to other molecules that the body needs)
- These include amino acids, which are used to build proteins such as **fibrinogen**, a protein found in blood plasma that is important in blood clotting
- **Excess amino acids** absorbed in the blood that are not needed to make proteins **cannot be stored**, so they are broken down in a process called **deamination**
  - The amino group of all amino acids - NH_2 (which contains the nitrogen atoms) is removed, hence the term de-**amin(o)**-ation
- Enzymes in the liver split up the amino acid molecules
- The part of the molecule which contains **carbon** is turned into **glycogen** and stored
- The other part, which contains **nitrogen**, is turned into **ammonia**, which is highly toxic, and so is immediately converted into **urea**, which is less toxic
- The urea dissolves in the blood and is taken to the **kidney** to be excreted
- A small amount is also excreted in **sweat**



Amino acid groups

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- The toxic consequences of high urea levels, if it is not excreted effectively, are very serious:
  - Cell death
  - Reduced response to insulin, leading to diabetes
  - Deposits inside blood vessels

Examiner Tips and Tricks

Excretion and egestion are two terms that often get confused:

Excretion is the **removal from the body of waste products** of metabolic reactions, toxic substances and substances in excess of requirements.

Egestion is the **expulsion of undigested food waste** from the anus.

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